

Target Table: `customer` Engine: PostgreSQL / ANSI SQL Total Queries: 10 Analytical + Schema Standardization

1. Schema Normalization & Standardization

Standardizing legacy column names with spaces and special characters to snake_case format for optimal querying and indexing efficiency.

Table Column Renaming Script

DDL / Maintenance

```
-- ALTER TABLE Statement Series
ALTER TABLE customer RENAME COLUMN "Customer ID" TO customer_id;
ALTER TABLE customer RENAME COLUMN "Age" TO age;
ALTER TABLE customer RENAME COLUMN "Gender" TO gender;
ALTER TABLE customer RENAME COLUMN "Item Purchased" TO item_purchased;
ALTER TABLE customer RENAME COLUMN "Category" TO category;
ALTER TABLE customer RENAME COLUMN "Purchase Amount (USD)" TO purchase_amount;
ALTER TABLE customer RENAME COLUMN "Location" TO location;
ALTER TABLE customer RENAME COLUMN "Size" TO size;
ALTER TABLE customer RENAME COLUMN "Color" TO color;
ALTER TABLE customer RENAME COLUMN "Season" TO season;
ALTER TABLE customer RENAME COLUMN "Review Rating" TO review_rating;
ALTER TABLE customer RENAME COLUMN "Subscription Status" TO subscription_status;
ALTER TABLE customer RENAME COLUMN "Shipping Type" TO shipping_type;
ALTER TABLE customer RENAME COLUMN "Discount Applied" TO discount_applied;
ALTER TABLE customer RENAME COLUMN "Promo Code Used" TO promo_code_used;
ALTER TABLE customer RENAME COLUMN "Previous Purchases" TO previous_purchases;
ALTER TABLE customer RENAME COLUMN "Payment Method" TO payment_method;
ALTER TABLE customer RENAME COLUMN "Frequency of Purchases" TO frequency_of_purchases;
```

2. Executive Data Analysis & Insights Queries

Q1: Total Revenue Generated by Gender

Revenue Breakdown

```
SELECT gender, SUM(purchase_amount_usd) AS revenue
FROM customer
GROUP BY gender;
```

Q2: Discount Users Spending Above Average

Customer Behavior

```
SELECT customer_id, purchase_amount_usd
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount_usd >= (SELECT AVG(purchase_amount_usd) FROM customer);
```

Q3: Top 5 Products by Average Review Rating

Product Quality

```
SELECT item_purchased, ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

Q4: Average Spend Comparison — Standard vs. Express Shipping

Logistics Analytics

```
SELECT shipping_type, ROUND(AVG(purchase_amount_usd), 2)
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Q5: Impact of Subscription Status on Spend & Revenue

Subscription Analysis

```
SELECT subscription_status,
       COUNT(customer_id) AS total_customer,
       ROUND(AVG(purchase_amount_usd), 2) AS avg_spend,
       ROUND(SUM(purchase_amount_usd), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY avg_spend, total_revenue;
```

Q6: Top 5 Products with Highest Discount Penetration Rate

Promotion Effectiveness

```
SELECT item_purchased,
       ROUND(100.0 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Q7: Customer Segmentation (New, Returning, Loyal)

Customer Lifetime Value

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
         CASE
           WHEN previous_purchases = 1 THEN 'New'
           WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
           ELSE 'Loyal'
         END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS "Number of Customers"
FROM customer_type
GROUP BY customer_segment;
```

Q8: Top 3 Most Purchased Products Per Category[Window Functions / Ranking](#)

```
WITH item_counts AS (  
    SELECT category, item_purchased,  
           COUNT(customer_id) AS total_orders,  
           ROW_NUMBER() OVER(PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank  
    FROM customer  
    GROUP BY category, item_purchased  
)  
SELECT item_rank, total_orders, category, item_purchased  
FROM item_counts  
WHERE item_rank <= 3;
```

Q9: Subscription Preference Among Repeat Buyers (>5 Purchases)[Retention Analysis](#)

```
SELECT subscription_status,  
       COUNT(customer_id) AS repeat_buyers  
FROM customer  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

Q10: Revenue Contribution by Age Group[Demographic Segmentation](#)

```
SELECT age_group,  
       SUM(purchase_amount) AS total_revenue  
FROM customer  
GROUP BY age_group  
ORDER BY total_revenue DESC;
```